Bangladesh University of Engineering and Technology

CSE 306

Computer Architecture Sessional

Assignment-1: 4-bit ALU

Section – B2

Group – 03

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A. Introduction

A processor unit implements the operations in a digital system or a digital computer. It is comprised of a number of registers and the digital functions implementing arithmetic, logic, shift and transfer micro-operations. A Central Processing Unit or CPU is basically the processor unit combined with a control unit.

An operation may be implemented in a processor unit either with a single micro-operation or with a sequence of micro-operations. The digital function that implements the micro-operations on the information stored in processor registers is called an Arithmetic Logic Unit or ALU. The control routes the source information from registers into the inputs of ALU. Then the ALU per- forms a given operation as specified by the control.

An ALU is a multi-operation, combinational-logic digital function that can perform a set of basic arithmetic and logic operations. It has a number of selection lines to select a particular operation (arithmetic or logic) in the unit. The selection lines are decoded within the ALU so that k selection variables can specify up to 2 distinct operations.

A parallel adder is the core of the arithmetic part of the ALU. Multiplexed inputs to the IC paves the way to achieve expected results for varieties of arithmetic operations. Besides, logical operations can be performed with their respective ICs, and in some efficient designs, a particular IC is used to do a different operation other than its intended one. There are also 4 status outputs (flags) in our designed ALU. They are denoted by C(Carry Flag), Z(Zero Flag), V(Overflow Flag), S(Sign Flag). Their representations carry out the following meanings:

CF: CF is simply the Cout of the adder used in the ALU. So, any carry out results in it being 1, otherwise it is set to 0. Logical operations cause it to be 0.

VF: If after any arithmetic operation, two positive numbers result in a negative output, or two negative numbers result in a positive output, that is an overflow and VF is set. After any logical operation, it is cleared.

SF: It reflects the output MSB.

ZF: If the last operation of the ALU yielded an output of 0, ZF is set, else it is 0.

B. Problem Specification with assigned instructions:

Design a 4-bit ALU with three selection bits CS2, CS1, CS0 (or S2, S1, S0) for performing the following operations:

Cin **S2**	**S1**	**S0**	**Function**
X	0	0	Decrement A
0	0	1	Transfer A
X	1	0	OR
0	1	1	Subtract with borrow
1	0	1	Subtract
1	1	1	Complement A

C. Design Steps with details:

For Cin:

s1s0 / **S2**	**00**	**01**	**11**	**10**
0	0	1	0	1
1	0	1	1	1

Cin=(s1)'s0 + s1(s0)' + s2s1

= (s1 XOR s0) + s2s1

For Yi:

s0Bi / **s2s1**	**00**	**01**	**11**	**10**
00	1	1	1	1
01	1	1	0	1
11	1	1	1	1
10	1	1	0	1

Yi = (s0)’ + (s2)’(s1)’ + s2s1 + (Bi)’

=(s0)’ + (s1 XOR s2)’ + (Bi)’

= (s0)’ + ((s1 XOR s2). Bi)’

Using a mux-

s0	**Yi**
0	1
1	((s1 XOR s2). Bi)’

For Xi:

Xi = [Ai ((s0)’ + (s1)’ + (s2)’)]

+ Bi s1 (s0)’

+ (Ai)’ s2s1s0

Using a mux-

s0	**Xi**
0	Ai + s1Bi
1	Ai ((s1)’ + (s2)’) + (Ai)’ s1s2 = Ai (s1s2)’ + (Ai)’ (s1s2) =A XOR (s1s2)

D. Truth Table:

Control Signals			Xi	Yi	Cin=S2	Output(F)	Function Name
CS2	**CS1**	**CS0**					
X	0	0	Ai	1 (all 1's)	0	A-1 = A +(all 1's)	Decrement A
0	0	1	Ai	1 (all 1's)	1	A	Transfer A
X	1	0	Ai + Bi	1 (all 1's)	1	A + B	OR
0	1	1	Ai	(Bi)'	0	A-B-1 = A + B'	Subtract with borrow
1	0	1	Ai	(Bi)'	1	A-B = A+B'+1	Subtract
1	1	1	(Ai)'	1 (all 1's)	1	A'	Complement A

E. Block Diagram

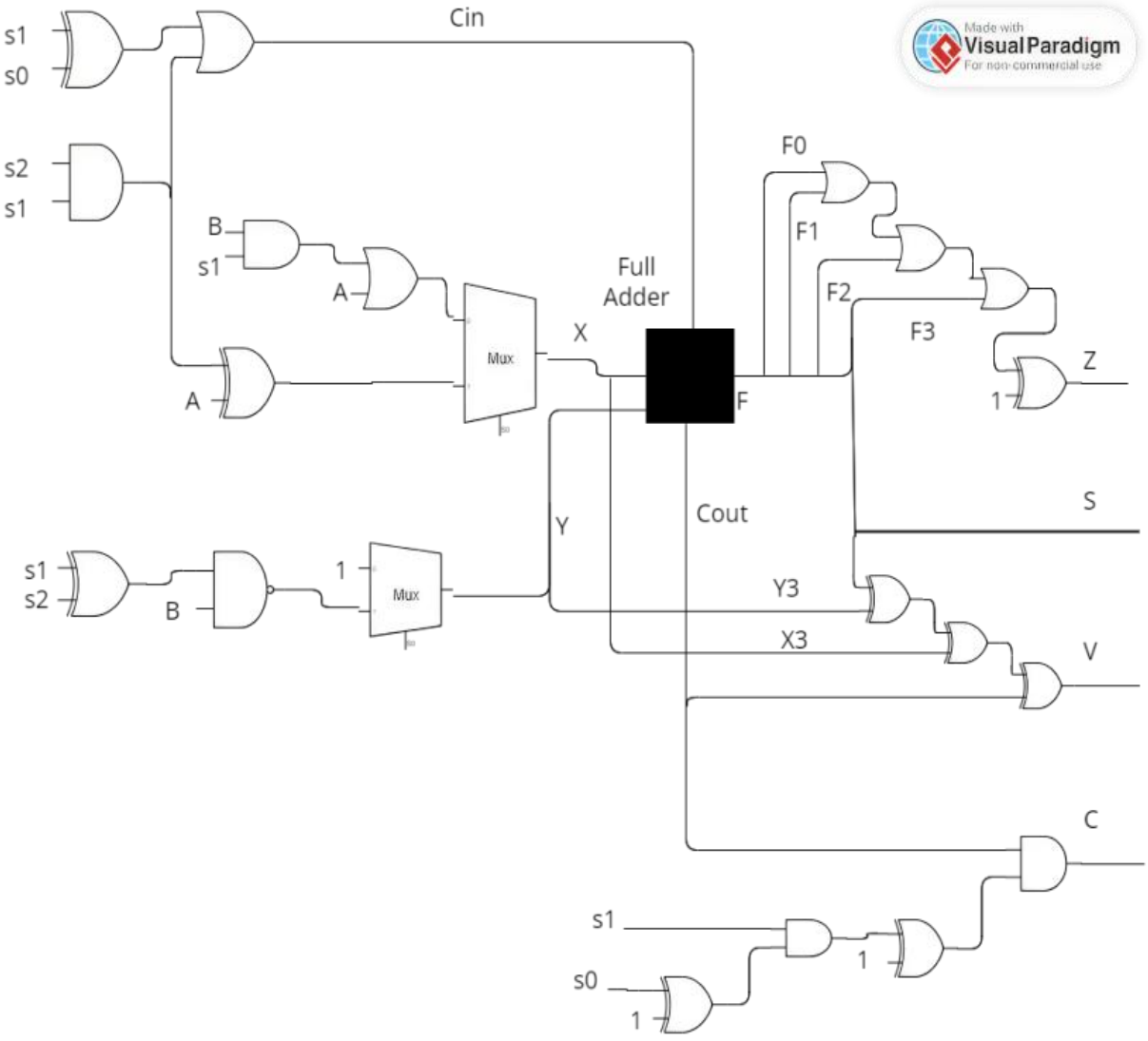


Fig: block diagram of ALU

F. Complete Circuit Diagram:

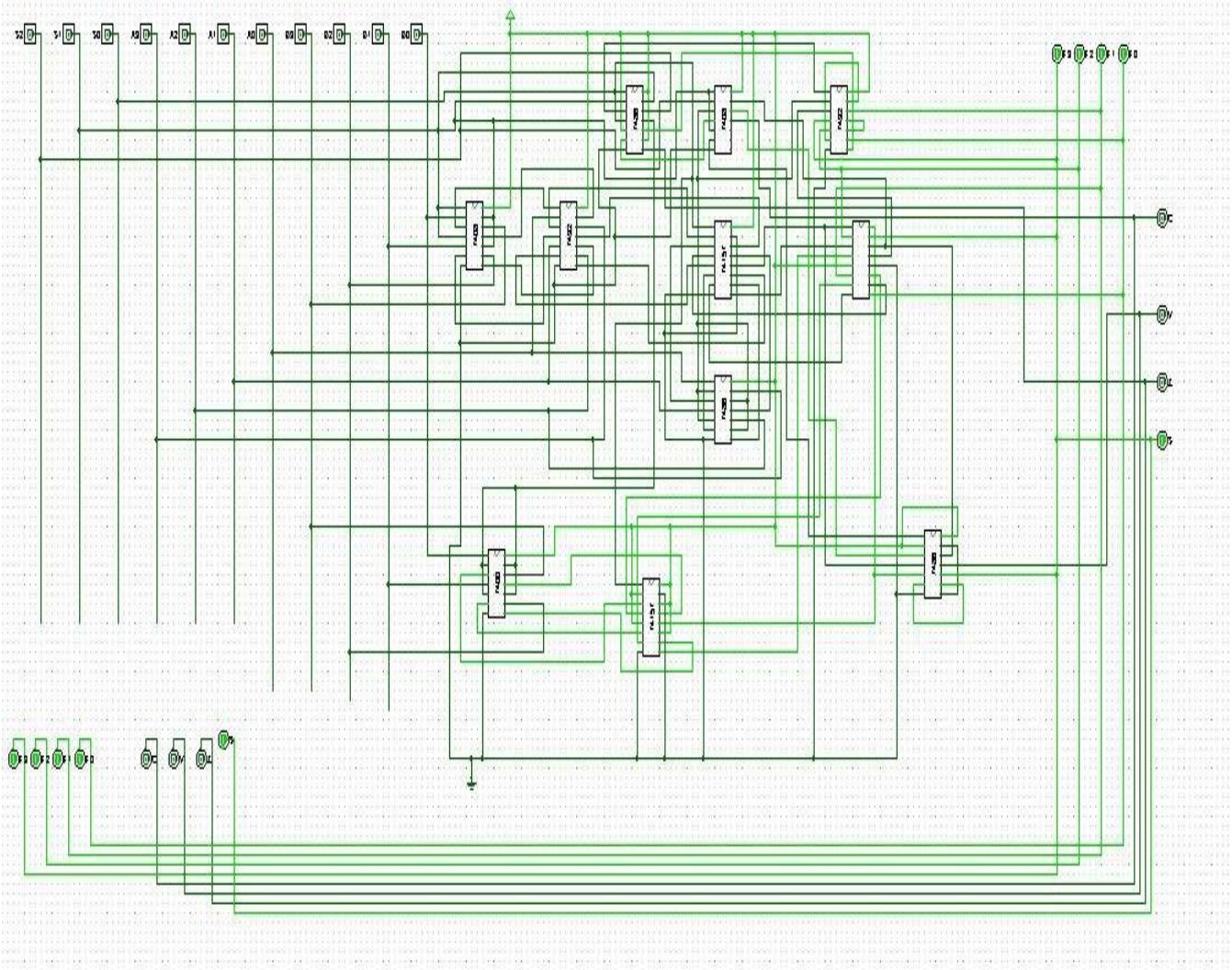


Fig: The complete circuit diagram using logisim

G. ICs used with count as a chart

IC	QUANTITY
IC 7400	1
IC 74157	2
IC 7483	1
IC 7432	2
IC 7408	2
IC 7486	3
Total	11

Table 3: ICs Used with quantity

H. The simulator used along with the version number

Logisim –2.7.1

I. Discussions

The task of using a 4-bit arithmetic logic unit (ALU) with a limited number of ICs presented challenges and required optimized design. We had to use mux for optimization. The hardware implementation also presented challenges such as the planned installation of modules. Proper grounding of all switches and breadboards is critical to the stability and reliability of the circuit. We also noted that because of electrical consumption considerations, choosing a green bulb over a red bulb means that electricity is more efficient. Also because of having an AC adapter, at the last time all the bulbs had stopped working. Having solved various challenges, we are optimistic that our 4-bit Arithmetic and Logic Unit (ALU) implementation will be successful. Our focus on logical optimizations and efficient hardware design aimed at reducing the number of ICs has hopefully resulted in a practical and flexible ALU.

J. Contribution:

1. 2005114 finalised the design with minimization.

2.2005095 also gave a primary design.

3.2005114, 2005096, 2005111 completed the Logisim part, 2005119 checked possible input, output from Logisim.

4.2005096 and 2005119 bought all the necessary components for the project.

5.2005114 and 2005095 has done half hardware part, 2005119 and 2005096 has done other half part, switching and some other necessary connections were done by 2005111.

6.Report has been written by 2005095, 2005096, 2005111, 2005114, 2005119 combinedly.